



Effects of a Chinese herbal medicine mixture on growth performance, immune response, and gene expression in Chinese soft-shelled turtles (*Pelodiscus sinensis*)

Mengyan Lan^{a,1}, Yaguang Wang^{a,1}, Yuqi Mu^a, Yali Li^a, Zhao Zhang^{a,*}, Yueqiang Guan^{a,b,**}

^a College of Life Sciences, Hebei University, Baoding 071002, China

^b Innovation Center for Bioengineering and Biotechnology of Hebei Province, Baoding 071002, China

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ABSTRACT

Chinese herbal medicines (CHMs), known for their safety, low toxicity, and minimal impact on the environment, are often used as a natural feed additive in aquaculture to prevent or control diseases of aquatic animals. However, their effect on cell responses of Chinese soft-shelled turtle (*Pelodiscus sinensis*) remains unexplored. In this study, the extract mixed with four Chinese herbs of *Astragalus membranaceus*, *Lycium barbarum*, *Rhizoma polygonati* and *Codonopsis pilosula* in proportions (7:8:3:2) was used as a feed additive for 60 days to study their effects on the growth performance and immune response of *P. sinensis* in greenhouse mode. Results showed that the weight gain rate and the specific growth rate in the CHMs group were 46.63 % and 0.64 %, which were 1.13- and 1.15-fold of that in the control group, respectively. Besides, CHMs could significantly improve the survival rate by 76.94 % after artificial infection of *Aeromonas hydrophila*. The CHMs group also exhibited significant increases in the activities of total superoxide dismutase (27.76 %), catalase (41.28 %), glutathione peroxidase (14.62 %), and glutathione sulfur transferase (36.75 %) compared to the control group. Conversely, malondialdehyde concentration was reduced to 67.82 % of that in the control group. Moreover, the concentration of lysozyme and complement C3, as well as the activity of acid phosphatase, rose by 16.43 %, 24.30 %, and 27.68 %, respectively. Comparative transcriptome analysis revealed that growth, immunity, and lipid metabolism-related pathways are involved in the response to CHMs. Among these, the two pathways of the cell cycle and the cytokine-cytokine receptor interaction may play important roles in the growth and immunity of *P. sinensis*. SMOX, STAT3, and HMGCS1 may serve as key genes in CHMs-stimulated immune response in *P. sinensis*; GADD45B, ARG1, ACVR1C, and BMP7 may be crucial genes in promoting the growth of *P. sinensis*. This study provides a theoretical and practical basis for the green and efficient culture of *P. sinensis*.

* Corresponding author.

** Corresponding author at: College of Life Sciences, Hebei University, Baoding 071002, China.

E-mail addresses: zhangzhao900626@sina.com (Z. Zhang), guan Yueqiang@hbu.edu.cn (Y. Guan).

¹ These two authors contributed equally to this work.

1. Introduction

Chinese soft-shelled turtle (*Pelodiscus sinensis*), which has both edible and medicinal value, is widely cultivated as an economic species (Wang et al., 2013). Recent research has highlighted the rich content of essential amino acids (EAA) and polyunsaturated fatty acids (PUFA) in *P. sinensis* (Wang et al., 2021a). Consequently, enhancing the weight gain and immune response of this species is a critical concern for both farmers and scientific researchers.

Growth is a fundamental trait in selective breeding for *P. sinensis* (Chavanne et al., 2016; Zeng et al., 2021). Various strategies have been employed to increase the growth rate, including the supplementation of dietary feed additives such as housefly larvae, sodium butyrate, and brown fishmeal, which have been shown to improve growth (Fan et al., 2021; Ge et al., 2023; Zhong et al., 2024). Additionally, antioxidant capacity, measured through total antioxidant capacity (T-AOC) and total superoxide dismutase (T-SOD) activity, can influence growth by neutralizing oxygen free radicals (Wang et al., 2008). Metabolic and metabolism-related genes have also been implicated in affecting the growth rate of *P. sinensis* (Zeng et al., 2021).

Disease resistance is another crucial trait in selective breeding for *P. sinensis* (Chavanne et al., 2016). Strengthening the immune system is vital for disease control. The innate and adaptive immune mechanisms of *P. sinensis* have been characterized. Upon pathogenic stimulation, the synthesis of cytokines and chemokines triggers an inflammatory response, recruiting macrophages and neutrophils to the site of infection (Ji et al., 2024). Following pathogen recognition, activated T-cell and B-cell differentiate into effector cells to eliminate the pathogen (Ji et al., 2024). Complement C3, interleukin-6 (IL-6), and lysozyme (LZM) are involved in both innate immune and adaptive immunity. *Aeromonas hydrophila*, a prevalent gram-negative bacterium, poses a significant threat to freshwater aquaculture species, causing various diseases and substantial economic losses. *A. hydrophila* infections have been linked to over 15 conditions, including keratitis, red-neck disease, septicemia, and furunculosis (Chung et al., 2017; Ji et al., 2024). Enhancing disease resistance and survival rates against *A. hydrophila* is therefore a high priority.

Chinese herbal medicines (CHMs), known as “green medicine”, have been utilized in China for millennia (Pu et al., 2017). *Astragalus membranaceus*, *Lycium barbarum*, *Rhizoma polygonati*, and *Codonopsis pilosula*, which are among the CHMs selected for this study, have shown promise in aquaculture. For instance, *L. barbarum* polysaccharide increased the specific growth rate of Nile tilapia (*Oreochromis niloticus*) by 0.55 % (Zhang et al., 2020). Another study found that CHMs significantly increased the weight gain and specific growth rate of Masu salmon (*Oncorhynchus masou*) (Wang et al., 2019). Besides, CHMs are used in treating various diseases in China, primarily through the bioactivity of active substances such as polysaccharides, flavonoids, tannins, and alkaloids (Pu et al., 2017). These active ingredients can improve antioxidant capacity and immunity. Studies have reported that *A. polysaccharides* can improve the intestinal microbes of grass carp (*Ctenopharyngodon idellus*), activate the body's immune response, and protect grass carp from bacterial infection (Shi et al., 2021). *L. barbarum* polysaccharides improved the non-specific immunity of Nile tilapia (*Oreochromis niloticus*), decreased the apoptosis rate, and promoted the growth and development of tilapia (Zhang et al., 2020). The compounds of *A. membranaceus* and *L. barbarum* had hepatoprotective and anti-stress bioactivities in juvenile Pacific white shrimp (*Litopenaeus vannamei*) (Prathomya et al., 2019). Therefore, CHMs have a certain resistance to various diseases that occur in intensive aquaculture. Despite these benefits, the impact of CHMs on the growth and immune response of *P. sinensis* has not been thoroughly investigated.

In this study, the extract mixed with four Chinese herbs (*A. membranaceus*, *L. barbarum*, *R. polygonati*, and *C. pilosula*), in which the main component is polysaccharide, was used as a feed additive at a concentration of 0.15 %. The growth performance and immune responses were assessed, and a comparative transcriptome analysis was conducted to elucidate the potential mechanisms underlying the effects of CHMs on *P. sinensis*. This research evaluates the efficacy of CHMs as feed additives and provides a reference for the healthy cultivation of *P. sinensis*.

2. Materials and methods

2.1. Animals, materials, and bacteria strain

A total of 120 *P. sinensis*, whose average initial size was 270.71 ± 24.45 g, were used in this study. They were provided by Hebei Kangtai Chinese Soft-shelled Turtle Co., Ltd., Luquan District, Shijiazhuang City, Hebei Province, China. The compound Chinese herbal medicine was composed of *A. membranaceus*, *L. barbarum*, *R. polygonati*, and *C. pilosula* in a proportion (7:8:3:2), and the mixed extract was obtained by the following approach. Ten times the amount of water was added (W/V), followed by a hot reflux extraction for 1.5 h, centrifuging, separation of the supernatant, adding 8 times the amount of water (W/V) to the residue, repeating a hot reflux extraction for 1 h, centrifuging, separating the supernatant, combining 2 times the continued filtrate, adding 10 % maltodextrin, and finally spray drying. This follows the methods of the Chinese herbal medicine innovation team of the modern agricultural industry technology system of Hebei Province. The basal diet adopts the feed of juvenile soft-shelled turtles provided by Zhejiang Jinjia Aquatic Feed Co., Ltd. The strain of *A. hydrophila* was donated by Li Quanzhen, a research scientist at Hebei Fisheries Technology Extension Station.

2.2. Feeding trial

The feed trial contained two duplicate treatments, the experimental group (EG) and the control group (CG). EG was fed with the basal diet plus 0.15 % compound Chinese herbal medicine extract, and CG was fed with the basal diet (Crude protein 38.86 %; Crude lipid 9.19 %; Crude ash 18.94 %; Gross energy 18.15 %; Calcium 3.32 %). The feeding trial site was provided by Hebei Kangtai Chinese Soft-shelled Turtle Breed Co., Ltd., and the turtles were cultured in a sunshade greenhouse. In the feeding trial, each group had two parallel 300 m² culture ponds with a depth of 0.7 m. In each pond, 1000 turtles were stocked. Before the test, the pond was cleaned and

the surrounding environment also was cleaned and fully disinfected. The experiment was started after 15 d of basal diet domestication. The water temperature was kept at 25–30 °C, changes in water temperature were recorded, and feeding was 2 %–3 % of body weight twice a d (8:00, 17:00). The number of morbidities and deaths was recorded, and the culture experiment lasted for 60 d.

2.3. Sample collection

The initial body weight and final body weight of each group of *P. sinensis* were statistically tested, and the weight gain rate (WGR), specific growth rate (SGR), survival rate, and challenge survival rate were calculated. The data formulas for each index were:

$$\text{Weight gain rate (WGR, \%)} = (W_t - W_0) / W_0 \times 100\%;$$

$$\text{Specific growth rate (SGR, \% / d)} = (\ln W_t - \ln W_0) / n \times 100\%;$$

$$\text{Survival rate (SR, \%)} = \text{final number} / \text{initial number} \times 100\%;$$

In the formula, W_t represents the final average body weight of soft-shelled turtles (g), W_0 represents the initial average weight of soft-shelled turtles (g), and n is the number of days of rearing (d).

Fifteen turtles were randomly collected from each pool for use in the challenge test. During the experiment, six turtles were collected from each pool on 0, 15, 30, 45, and 60 d for the detection of antioxidant system-related indicators and immune-related indicators. Meanwhile, the liver tissues of three turtles were collected in the experimental group and the control group for high-throughput transcriptome sequencing on the 15th d.

2.4. The bacterial challenge test

The turtle that had been reared for 60 d was used as the challenge object, and the turtle with the same body weight was selected and acclimated in constant-temperature aquarium in the laboratory for one week. The water temperature in the tank was controlled at 30 °C, the pH value was 7.25 ± 0.25 , and dissolved oxygen was greater than 2.0 ± 0.5 mg/L. *A. hydrophila* was inoculated from fresh slanted nutrient agar medium in eppendorf tubes supplemented with sterile nutrient broth, shaken, and cultured at about 28 °C on a constant temperature shaker for 24 h, and the activated bacterial solution was collected and further centrifuged. The bacteria were washed with sterilized physiological saline (PBS) more than three times, and then the concentration was adjusted to 1.0×10^9 cfu/mL with this PBS for use. The experimental *P. sinensis* was injected intraperitoneally at a dose of 1 µL/g body weight. The experiment was divided into control and experimental groups, the control group was the soft-shelled turtle that was fed with the control feed during the feeding period, and the experimental group was the turtle that was fed with the feed added with CHMs. Each group had three parallel tanks, and each tank was stocked with 10 turtles. During the challenge test, we observed and recorded survivorship for 7 d. The survival rate was calculated at the end of the experiment.

$$\text{Survival rate after challenge} = (1 - \text{death} / \text{total}) \times 100\%.$$

2.5. Antioxidant system analysis

The liver tissues of *P. sinensis* on 0, 15, 30, 45, and 60 d from each group were homogenized using a homogenizer (Bertin, France) by adding 9 vol (v/w) of PBS (0.86 %). The homogenate was then centrifuged at 4000 rpm (Biofuge Primo R, Thermo) for 10 min at 4 °C. Supernatants were immediately used for antioxidant status analysis. The antioxidant status was measured by total antioxidant capacity (T-AOC), total superoxide dismutase (T-SOD) activity, catalase (CAT) activity, glutathione peroxidase (GSH-Px) activity, glutathione S-transferase (GSH-ST) activity, malondialdehyde (MDA) content, and total protein concentration (TP). The results are expressed as units per milligram protein. This was determined using commercial assay kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China), according to the manufacturer's instructions.

2.6. Immunological indices analysis

The serum of *P. sinensis* was obtained on 0, 15, 30, 45, and 60 d from each group by centrifuging the blood after standing at 4000 rpm (Biofuge Primo R, Thermo) for 30 min at 4 °C. The immune status was measured by the activities of acid phosphatase (ACP) and alkaline phosphatase (AKP), the contents of complement C3, IL-6, and LZM. Among them, the measure of ACP, AKP activities, and LZM contents were used in commercial assay kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China) according to the manufacturer's instructions. The contents of complement C3 and IL-6 were determined by an enzyme-linked immunosorbent assay (ELISA, Shanghai Yuanmu Biotechnology Co., Ltd., Shanghai, China).

2.7. RNA extraction, cDNA library construction, and deep sequencing

Three turtles were selected from each of the two different treatment groups, the liver tissues were collected for RNA extraction, and the total RNA was extracted using TRIzol reagent (Invitrogen, Carlsbad, CA, USA). RNA concentration was determined using the

Qubit® RNA Assay Kit in a Qubit®2.0 Fluorometer (Life Technologies, CA, USA), and the integrity was assessed using the RNA Nano 6000 Assay Kit of the Bioanalyzer 2100 system (Agilent Technologies, CA, USA). Transcriptome sequencing was conducted by Novogene Technology Co., Ltd. (Beijing, China). A total of 3 µg RNA per sample was used for RNA sample preparation. Sequencing libraries were constructed using the NEBNext® Ultra™ RNA Library Prep Kit (Illumina, USA) following the manufacturer's recommendations, and index codes were added to attribute sequences to each sample. Clustering of the index-coded samples was performed on a cBot Cluster Generation System using TruSeq PE Cluster Kit v3-cBot-HS (Illumina, USA) according to the manufacturer's instructions. After cluster generation, the library preparations were sequenced on an Illumina HiSeq platform.

2.8. Quality control and reads mapping to the reference genome

Clean reads were obtained by removing low-quality reads and reads that contained adapter and ploy-N from raw reads. The Q20, Q30, and GC contents of the clean data were calculated. The reference genome was downloaded from the ensembl genome database website, <http://ftp.ensembl.org/pub/release-96/fasta/pelodiscus.sine.nsis/dna/>. The index of the reference genome was built, and paired-end clean reads were aligned to the reference genome using Hisat2 v2.0.5.

2.9. Screening and analysis of differentially expressed genes

Reads mapped to each gene were counted using featureCounts. Considering the effects of sequencing depth and gene length, gene expression levels were estimated using FPKM, which refers to the expected number of fragments per kilobase of transcript sequence fragments sequenced per megabase pair. Differential expression analysis was performed using DESeq2 (with biological replicates), and differentially expressed genes with $|\log_2(\text{FoldChange})| \geq 1$ and $\text{padj} \leq 0.05$ were screened out. Gene Ontology (GO) and KEGG pathway enrichment analyses of differentially expressed genes (DEGs) were implemented using the clusterProfiler R package.

2.10. Validation using quantitative real-time PCR

The RNA samples used for RNA sequencing (RNA-seq) were also used for quantitative real-time PCR (qRT-PCR) validation. Twelve DEGs were chosen to perform the qRT-PCR assay and 18 S rRNA was chosen as an internal reference gene. Primers (Table 1) for qRT-PCR were designed by Primer Premier 5.0. The qRT-PCR assays were conducted with TB Green® Premix Ex Taq™ (Tli RNaseH Plus) (TaKaRa, Japan) on the Bio-Rad Real-time PCR System (Bio-Rad, USA). The PCR program was 95 °C for 3 min, then 35 cycles of 95 °C for 10 s, annealing temperature (Tm) for 10 s, and 72 °C for 15 s. The $2^{-\Delta\Delta CT}$ method was used to calculate the fold changes in the expression level of chosen genes.

Table 1
Primers for qRT-PCR.

Primer name	Primer sequences 5'-3'	Product size / bp	TM/°C
ARG1-F	GGGATTGGGAAGGTGATG	126	58.3
ARG1-R	AAC TGGGGTGCCTGTAGC		
BNIP3-F	ATGACCTGTGCTGTTTGTTTC	70	61
BNIP3-R	TTCAGTATCTCTCTCCCCC		
CCND1-F	TGGAGACCATCAGAAGAGC	253	55.2
CCND1-R	AGGGGTTCAAACGATAAAA		
CPT1A-F	TGACCTGCGAATGAGCCA	156	60
CPT1A-R	GACATCACCCCAACCACA		
HMGCS1-F	ATTTTGTGAGAGAGGACTGCG	182	57.3
HMGCS1-R	CCTTCTTTTGCCACTGAGC		
LPL-F	ACATCTGCTACTTGGTCCCT	110	61
LPL-R	ATACATCCCTGTTACCGTCC		
MGST2-F	ATGGCTGGCGATTTAGTTT	111	60
MGST2-R	GGGCATGATCTTGTGCTGT		
ODC1-F	CTACTTCTGGAGCGGGCAA	125	60
ODC1-R	TCAAAGACACAACGGGCAT		
STAT3-F	CAGTTTACCACCAAGTCAG	142	57.3
STAT3-R	TGTTCTGTGCCCCAAGATGTTA		
SMOX-F	CAGGTGATGTTTTCGGGGGA	100	59
SMOX-R	TTTCAATGAGGTGAGCAGCC		
ACSL1-F	CCCACCAACGGATGAAAAAG	112	59
ACSL1-R	AGATGTCCCTGGAAGCAGG		
MAOA-F	GCTAAGGAAGCACGGGAGATT	200	60
MAOA-R	TTGCCAAGAGCGGTGGATT		
18 S rRNA-F	AAAGGAATTGACGGAAGGGCAC	162	55
18 S rRNA-R	GCTCCACCAACTAAGAACGG		

2.11. Statistical analysis

The value of each variable was calculated and expressed as mean $\pm$ standard deviation ($X \pm SD$). Statistical analysis was performed using the independent samples t-test (SPSS Ver 19.0).

3. Results and discussion

3.1. The effects of Chinese herbal medicines on the growth of *P. sinensis*

The weight gain rate and specific growth rate were measured and calculated after feeding *P. sinensis* with CHMs as a feed additive for 60 d. The results showed that the survival rate of both the experimental and control groups was above 99 % and the average final weight of the test group was 397 g, which was significantly higher than that of the control group ($P < 0.01$). The weight gain rate and the specific growth rate of the CHMs group were 46.63 % and 0.64 %, which were 1.13- and 1.15-fold of that in the control group, respectively (Table 2). In terms of weight gain, studies have shown that the polysaccharide-rich *Astragalus* significantly improves the growth performance of piglets, and *L. barbarum* extract can also improve the weight gain rate and feed conversion rate of grass carp and Nile tilapia (Huang et al., 2014; Mo et al., 2016). There are also studies reporting that *Codonopsis* has a growth-promoting effect by regulating hormone production and digestion (Khaki et al., 2014). In addition, CHMs improve the immunity of farmed animals, reduce the energy supply in terms of disease resistance, and supply more energy for growth, which in turn affects the changes in growth performance (Wang et al., 2020). Therefore, it is reasonable to find positive effects of CHMs on the growth of *P. sinensis*.

3.2. The immunoprotective effect of Chinese herbal medicine against *Aeromonas hydrophila* infection

A 7-day *A. hydrophila* challenge test was conducted on *P. sinensis* after 60 d of feeding. The results showed that the survival rate of the experimental group was significantly higher ($P < 0.01$). The survival rate of the CHMs group was 76.67 %, whereas it was only 43.33 % for the control group (Table 3). Several studies have shown that CHMs can improve the resistance of aquatic animals to invasion by *A. hydrophila*. A mixed extract of *Astragalus*, *Crataegus pinnatifida*, *Glycyrrhiza uralensis*, and *Lonicera japonica* has been reported to improve the antibacterial activity of Nile tilapia against *A. hydrophila* (Tang et al., 2014). The mixed extract of *Astragalus radix* and *Ganoderma lucidum* could enhance the immune response of carp (*Cyprinus carpio*), improve the phagocytosis of leukocytes in the organism, increase the content of LZM, and improve the rate of immune protection against *A. hydrophila* in carp (Yin et al., 2009). This is also consistent with our results that the survival rate of the CHMs group was significantly higher than that of the control group in the challenge test of this study. These results also suggested that the CHMs used in the present study can improve immunity and enhance the disease resistance of *P. sinensis*.

3.3. Effects of Chinese herbal medicines on the antioxidant system

In aquaculture, improving antioxidant capacity becomes critical as both the growth and the development of animals could be affected by excess free radicals (Chang et al., 2009). T-AOC and T-SOD activities are often used as evaluation parameters of antioxidant capacity (Wang et al., 2008). T-AOC, as the name suggests, indicates the total level of the antioxidants and antioxidative enzymes in the organism. T-SOD is at the forefront of resistance to oxidative stress, followed by CAT and GSH-Px, which work in concert to deal with free radicals. The three kinds of antioxidative enzymes respond to the organic ability to scavenge free radicals (Sun et al., 2014). In this experiment, T-AOC significantly increased at both 45 and 60 d, including by 24.39 % at 60 d (Fig. 1e). The T-SOD activity in the experimental group significantly increased at 45 and 60 d, with an increase of 27.76 % at 60 d (Fig. 1a). CAT activity significantly increased at 30, 45, and 60 d, with an increase of 41.28 % at 60 d (Fig. 1b). GSH-Px activity significantly increased at 60 d, with an increase of 14.62 % (Fig. 1c). GSH-ST activity significantly increased at 45 and 60 d, with an increase of 36.75 % at 60 d (Fig. 1d). The active components of *A. membranaceus* and *R. polygonati*, including polysaccharides, methyl glycosides, polyphenols, and flavonoids, were found to have antioxidant effects (Han, 2016; Zhang et al., 2019). It was found that both *L. barbarum* and *C. pilosula* can enhance the activities of antioxidative enzymes such as SOD, CAT, and GSH-Px (Shen et al., 2021a; Zhou et al., 2021). These studies are consistent with the results in Fig. 1 in the present study.

Correspondingly, MDA can be used to evaluate the degree of lipid peroxidation (Marnett et al., 1999). In this case, it was

Table 2
Growth performance of *Pelodiscus sinensis* for different treatments within 60 d.

	EG	CG
Survival rate (SR, %)	99.64 $\pm$ 0.06	99.58 $\pm$ 0.04
Initial weight (g)	270.75 $\pm$ 23.56	270.67 $\pm$ 26.35
Final weight (g)	397.00 $\pm$ 30.54**	362.25 $\pm$ 25.63
Weight gain rate (WGR, %)	46.63 $\pm$ 2.45	33.84 $\pm$ 1.10
Specific growth rate (SGR, %)	0.64 $\pm$ 0.06	0.49 $\pm$ 0.02

Note: EG: the experimental group, CG: the control group. ** represents extremely significant difference, $p < 0.01$; * represents significant difference, $p < 0.05$. Data are presented as mean $\pm$ standard deviation ($n = 2$).

Table 3Survival rate of different treatments within 7 d of *Pelodiscus sinensis*.

Treat-ments	Parallel samples	Number of tested turtles	Average weight (g)	Mortality (ind.)	Survival rate (%)	Average survival (%)
EG	I	10	365.60 ± 23.43	2	80.00	76.67**
	II	10	368.00 ± 38.03	3	70.00	
	III	10	364.50 ± 35.06	2	80.00	
CG	I	10	361.80 ± 24.28	5	50.00	43.33
	II	10	362.40 ± 22.04	6	40.00	
	III	10	344.80 ± 27.17	6	40.00	

significantly decreased at 30, 45, and 60 d, and was only 67.82 % of the control at 60 d (Fig. 1f). Previous research has reported that the polysaccharide in *L. barbarum* and *P. sibiricum* could inhibit hepatic oxidative damage in mice on a high-fat diet and was able to reduce the MDA content (Wu et al., 2010). This is consistent with the significant decrease in MDA after 30 d in the present study. In conclusion, CHMs improved the antioxidant capacity and attenuated the negative effects caused by oxidative stress of *P. sinensis*.

3.4. Effects of Chinese herbal medicine on immune responses

In addition to affecting the turtles' antioxidant system, CHMs also affect immune responses. LZM is a non-specific immune protein that plays a key role in the innate immune response (Chen et al., 2018). LZM levels in the experimental group significantly increased at 30, 45, and 60 d compared to the control group, with an increase of 16.43 % at 60 d (Fig. 2a). Studies have shown that *A. membranaceus* and *L. barbarum* polysaccharides have obvious effects on increasing the LZM content in the serum of grass carp (*Ctenopharyngodon idellus*) and golden pompano (*Trachinotus ovatus*) (Li et al., 2019). This is consistent with the results of the present experiment.

The complement system is one of the important immune systems in aquatic animals and the complement plays an important role in the regulation and enhancement of humoral immunity (Nakao et al., 2011). In the process of complement activation, both complement C3 and complement C4 play a regulatory and phagocytic role, producing more C3 molecules—C3 plays a greater role than C4 (Aderem et al., 1999). Compared to the control group, the complement C3 concentration in the CHMs group increased by 24.3 % at 60 d (Fig. 2b). Some studies reported that the addition of 0.1 % *Astragalus* polysaccharide and *L. barbarum* polysaccharide to the diet can increase the C3 content in the serum of Nile tilapia and Northern Snakehead *Channa argus* (Zhu et al., 2021). It is thus hypothesized that in the present experiment, the level of complement C3 was significantly increased at 60 d, which may be related to the extracts of *A. membranaceus* and *L. barbarum* contained in the formula.

ACP and AKP are important metabolic enzymes in aquatic animals. ACP is a marker enzyme for lysosomes in macrophages (Tanaka et al., 1990). AKP can directly participate in phosphorus metabolism and maintain calcium and phosphorus balance (Butterworth., 1983). ACP activity significantly increased at 15, 30, 45, and 60 d, with an increase of 27.68 % at 60 d (Fig. 2c), whereas AKP activity did not significantly change throughout the test period (Fig. 2d). *L. barbarum* can significantly improve the NO synthesis, phagocytic ability, and ACP activity of macrophages (Gong et al., 2018). In another study, the addition of 0.25 %, 0.50 %, and 0.75 % *Astragalus* polysaccharides to the diet of Chinese soft-shelled turtle significantly increased the activity of ACP and AKP. However, supplementation of 1 % *Astragalus* polysaccharides in the diet decreased the activity of ACP, while the activity of AKP continued to increase (Yang et al., 2018). It is hypothesized that the lack of significant changes in AKP may be related to the concentration.

IL-6 is an important indicator for evaluating the immune status of aquatic animals, being involved in the regulation of the cellular immune system (Kim et al., 2017). IL-6 concentration slightly decreased at 60 d, 1.65 % of the control group (Fig. 2e). Increased levels of IL-6 are associated with inflammation (Sushaket al., 2020). *Polygonatum* polysaccharides can reduce inflammatory factors such as TNF- α , IL-6, IL-1 β , and iNOS (Liu et al., 2021). The effect of CHMs on the expression of IL-6 in *P. sinensis* needs further research.

In sum, the effect of CHMs on LZM and ACP in the serum of *P. sinensis* was more pronounced in the experimental group, and LZM and ACP activities were significantly increased in the experimental group. With the extension of feeding time, the complement C3 level significantly increased and the IL-6 level significantly decreased.

3.5. Transcriptome sequencing and identification of differentially expressed genes

In the study, cDNA libraries were constructed from mRNAs isolated from the experimental groups (EG1, EG2, and EG3) and control groups (CG1, CG2, and CG3). In sum, 279,390,512 raw reads and 273,268,966 clean reads were obtained. The error rate of base sequencing was 0.02 % for the six libraries. Detailed information is provided in Table S1. Sequence mapping was conducted using clean reads from the six constructed libraries, the total mapped ratio ranged from 81.86 % to 83.65 %, and the uniquely mapped ratio ranged from 77.78 % to 79.82 %. Detailed information on the mapping results is shown in Table S2. The DESeq2 R package was used to identify DEGs with an adjusted p-value < 0.05, and |log₂ FoldChange| > 1. A total of 921 DEGs were obtained from two groups, 478 up-regulated genes and 443 down-regulated genes were identified (Fig. 3). To confirm the veracity and reliability of the DEGs identified by RNA-seq, 12 genes related to growth and immunity were selected for qRT-PCR validation, and 18 S rRNA was used as the internal reference gene. Real-time quantitative PCR primers are shown in Table 1. Six of them were up-regulated, while the other six were down-regulated. The expression patterns of the 12 genes of qRT-PCR results were consistent with the RNA-seq results (Fig. 4), which suggested that the RNA-seq results were robust.

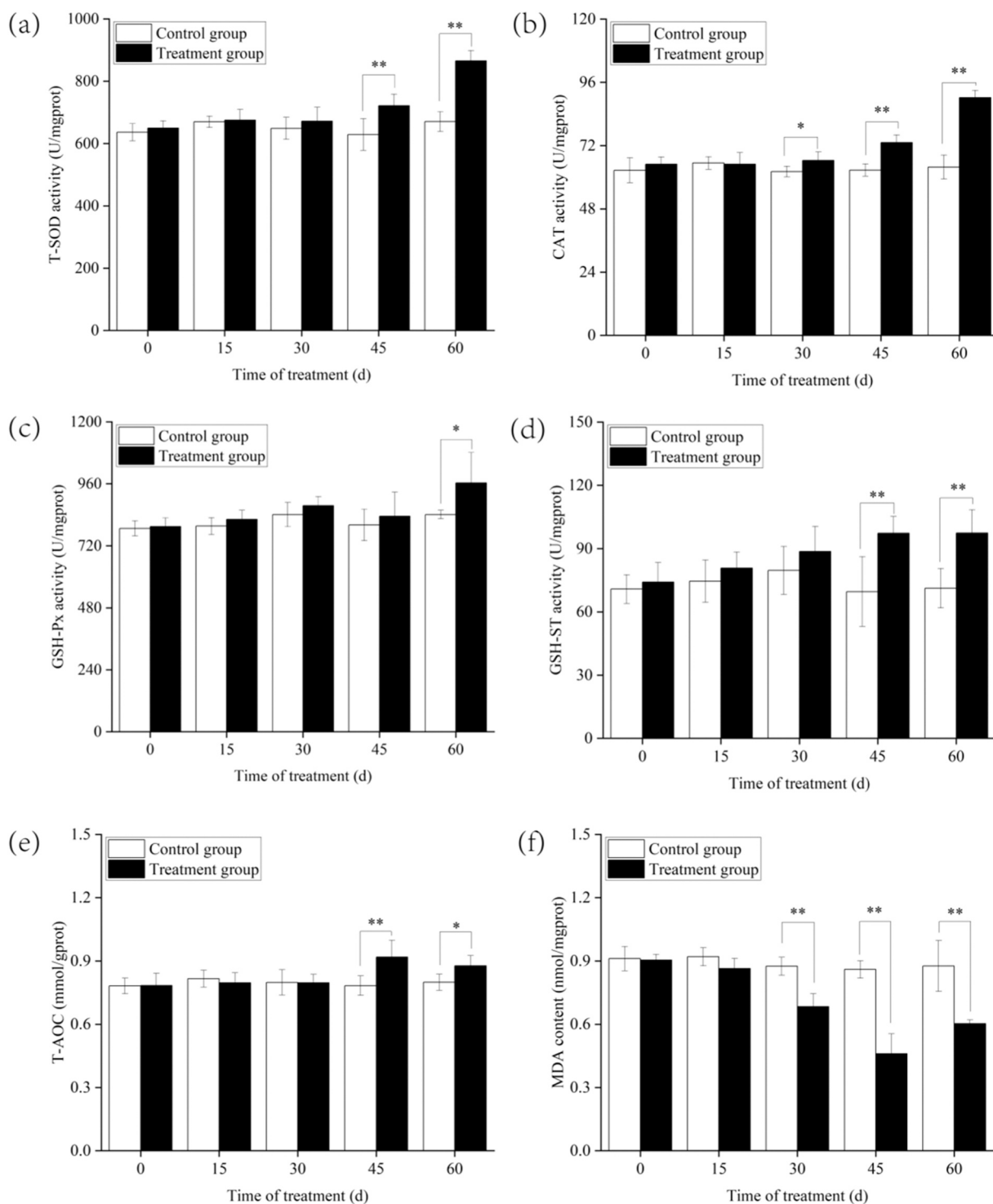


Fig. 1. The changes of the antioxidant system in *P. sinensis* after fed with the diet supplemented with Chinese herbal medicines. Differences in the control and treatment group for: (a) Total superoxide dismutase (T-SOD); (b) Catalase (CAT); (c) Glutathione peroxidase (GSH-Px); (d) Glutathione S-transferases (GSH-ST); (e) Total antioxidant capacity (T-AOC); (f) Malondialdehyde (MDA). * $p < 0.05$; ** $p < 0.01$. Data are presented as mean $\pm$ SD (n = 6).

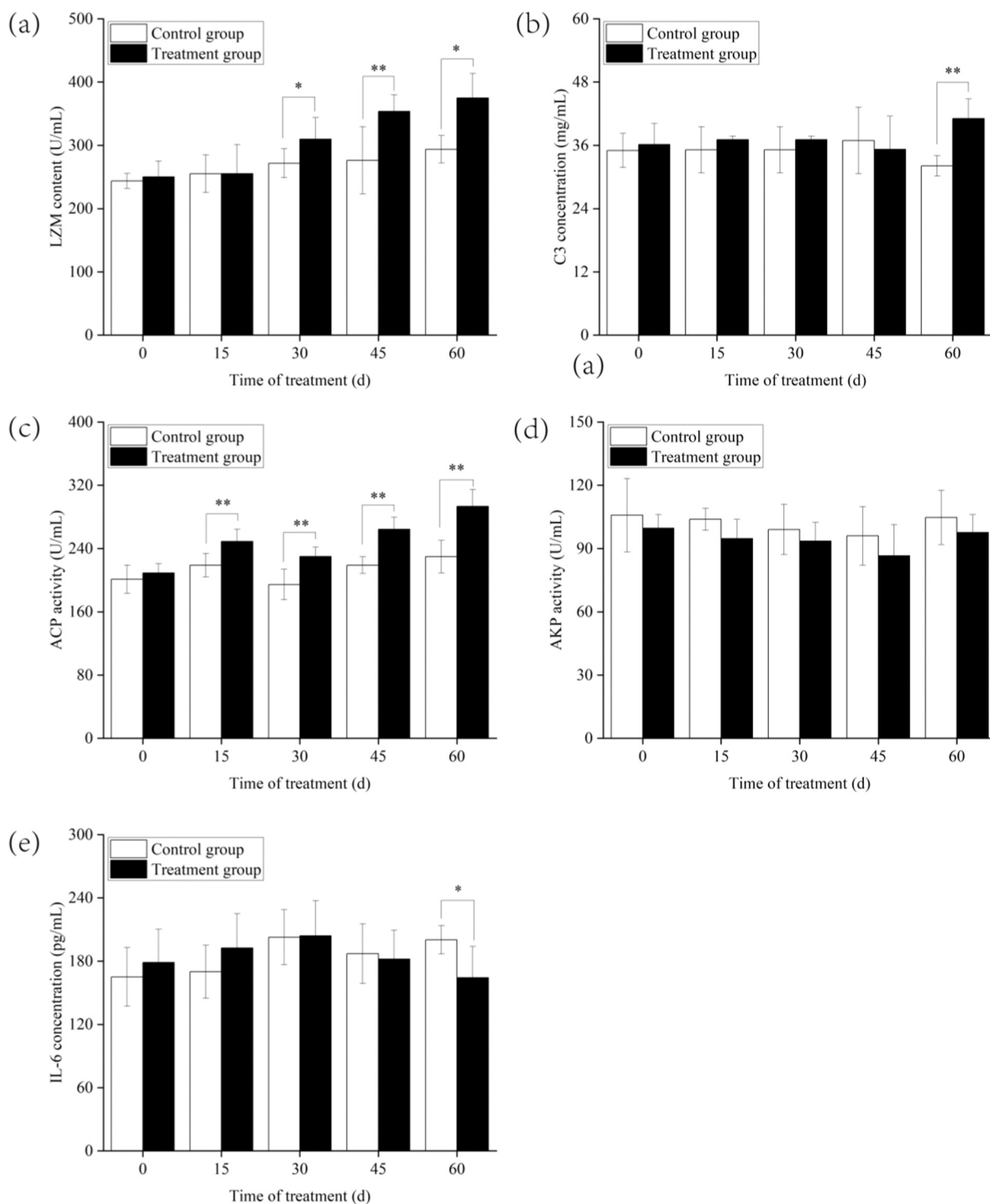


Fig. 2. The changes of immune-related indexes in *P. sinensis* after fed with the diet supplemented with Chinese herbal medicines. Differences in the control and treatment groups for: (a) Lysozyme (LZM); (b) Complement C3 (C3); (c) Alkaline phosphatase (ACP); (d) Acid phosphatase (AKP); (e) Interleukin-6 (IL-6). * $p < 0.05$; ** $p < 0.01$. Data are presented as mean $\pm$ SD ($n = 6$).

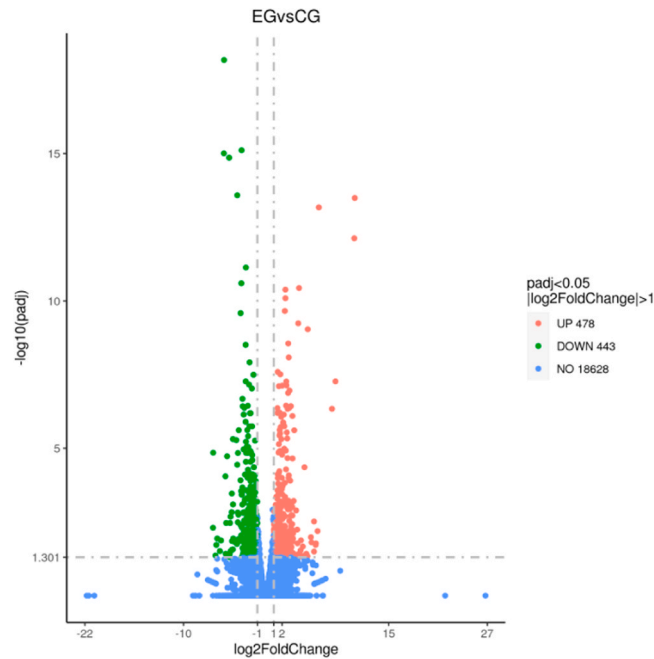


Fig. 3. Volcano Plot of DEGs distribution trends between the control and the experimental group.

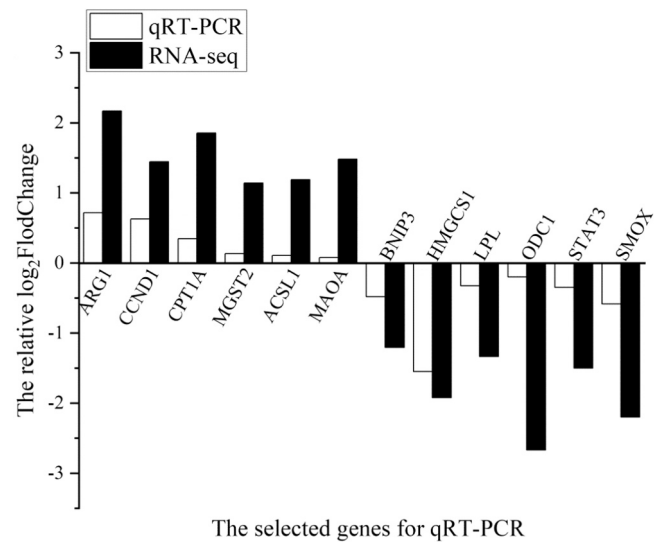


Fig. 4. Validation of quantitative real-time PCR. Comparison of the relative log₂ FoldChange between RNA sequencing and quantitative real-time PCR compared to the control, as normalized with the 18 S rRNA.

3.6. Molecular mechanisms of cellular responses under the influence of CHMs

3.6.1. KEGG enrichment analysis

The molecular mechanism of the cellular response after feeding *P. sinensis* with CHMs was explored. KEGG enrichment analysis was conducted, and a total of 138 pathways were enriched. The 20 most enriched pathways are shown in Fig. 5. Cell cycle (pss04110) was the most prominent pathway, which enriched the most differentially expressed genes. Among them, 13 significantly enriched pathways were divided into three categories related to the growth, immunity, and metabolism, as shown in Table 4 ($P < 0.05$). These pathways were enriched, effectively showing that CHMs significantly affected the growth of *P. sinensis*.

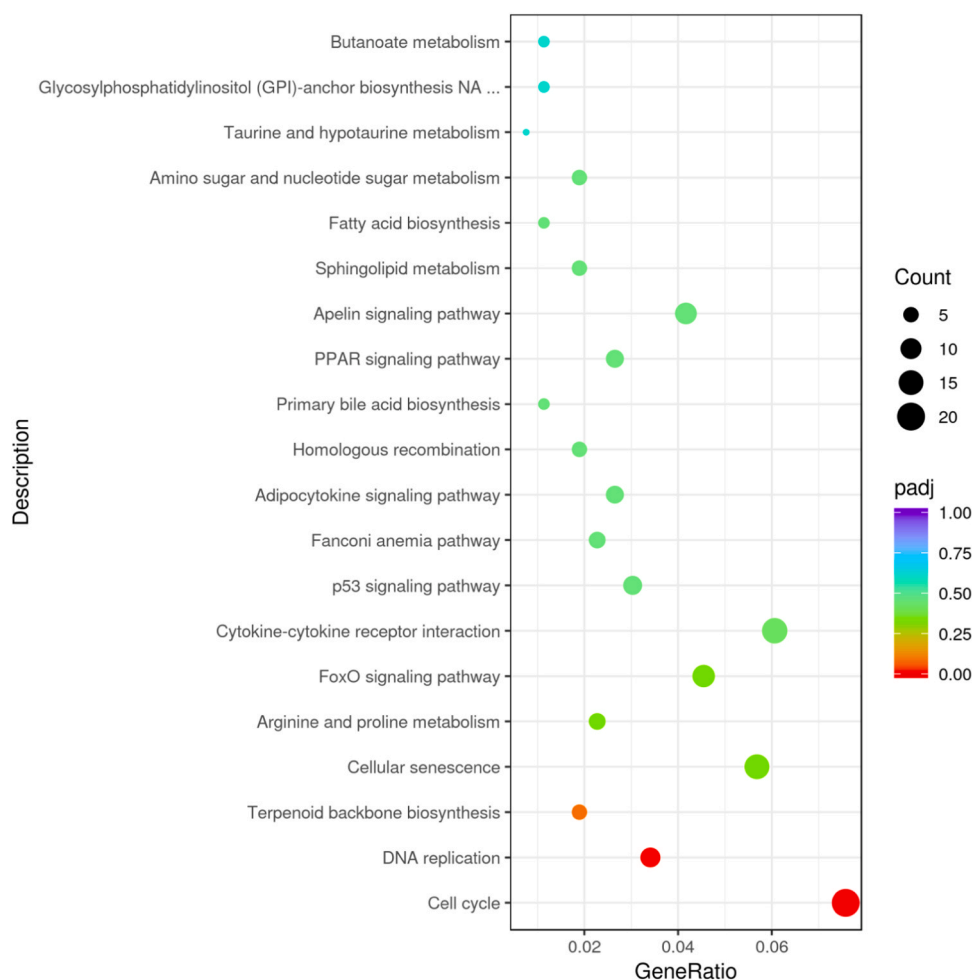


Fig. 5. KEGG classification of DEGs. The Y-axis represents the pathway name, the X-axis represents GeneRatio corresponding to each pathway, and the number of padj is represented by the color of dots. The number of differentially expressed genes under each pathway is indicated by the size of the dots.

3.6.2. Changes in the antioxidant system of *P. sinensis* under the influence of CHMs

Several studies have shown that CHMs can improve the antioxidant capacity of aquatic animals (Pu et al., 2017). In the antioxidant system, glutathione metabolism (pss00480) and peroxisome (pss04146) are two important pathways. Glutathione metabolism (pss00480) regulates the redox state of cells, balances oxidative stress in the organism, and plays an important role in the antioxidant system of aquatic animals (Cossu et al., 1997). While, the peroxisomes (pss04146) can dehydrogenate and break down unwanted organic compounds in cells to avoid toxicity due to accumulation in the organism (Di et al., 2023). Among them, are the chloride channel 5 gene (CLIC5), glutathione peroxidase 3 genes (GPX3), ENSPSIG00000003755 (glutathione S-transferase, N-terminal domain), superoxide dismutase 1 gene (SOD1), copper chaperone for superoxide dismutase gene (CCS) and ENSPSIG00000007847 (iron/manganese superoxide dismutases, C-terminal domain, iron/manganese SOD carbon-terminal domain) were up-regulated; glutathione S-transferase omega 1 gene (GSTO1) and spermine oxidase gene (SMOX) were down-regulated (Fig. 6A).

GSTO1 may enhance cell survival under stressful conditions through an antioxidant defense mechanism of elevated glutathione (Paul et al., 2015). It has been reported that GSTO1 is up-regulated in the face of nitrite stress in response to oxidative stress (Feng et al., 2021). The down-regulation of GSTO1 in this experiment may indicate that CHMs can reduce oxidative damage in *P. sinensis*. This is also consistent with the results of the antioxidant system index detection in this study (Fig. 1). In addition, it has been reported that the significant down-regulation of SMOX can inhibit the production of reactive oxygen species (ROS), and the accumulation of ROS can lead to changes in cellular redox status, which in turn triggers lipid peroxidation causing oxidative damage and an increase the MDA content (Cheng et al., 2020). Therefore, the inhibition of ROS affects the reduction of MDA content, which is also consistent with the results of MDA index determination in this study. In sum, CHM can increase antioxidant capacity by affecting some genes in glutathione metabolism (pss00480) and peroxisome (pss04146).

Table 4

The enriched KEGG pathway and related genes between the control group and experimental group.

KEGG ID	Description	p-value	Up genes	Down genes
Immune-related pathways				
pss04068	Foxo signaling pathway	0.01424	CCND1/CDK2	GADD45B/BNIP3/STAT3
pss03320	PPAR signaling pathway	0.02503	CPT1A/ACSL1/SLC27A5/CYP7A1	LPL/HMGCS1/ACSBG2
pss04060	Cytokine-cytokine receptor interaction	0.02116	BMP7/ACVR1C	IL1R2/CCL5/CCL16
Metabolism pathways				
pss00900	Terpenoid backbone biosynthesis	0.00175	MVK/PCYOX1L/ACAT2	HMGCS1
pss00330	Arginine and proline metabolism	0.01110	MAOA/ARG1	SMOX/ODC1/P4HA1/AMD1
pss04920	Adipocytokine signaling pathway	0.03488	CPT1A/ACSL1	STAT3/SLC2A1
pss00120	Primary bile acid biosynthesis	0.03998	CYP7A1/BAAT	
Growth-related pathways				
pss04110	Cell cycle	3.24E−06	MCM2/CCND1/MCM3/CDC25A/MCM5/MCM7/CHEK2/E2F1/ORC3/CDC45/CDC6/CDC20/CDK2	GADD45B
pss03030	DNA replication	5.04E−05	MCM2/MCM3/MCM5/POLE/MCM7/PRIM2	
pss04218	Cellular senescence	0.01067	CCND1/CDC25A/CHEK2/E2F1/CDK2	GADD45B/NFATC3
pss04115	p53 signaling pathway	0.02850	CCND1/CHEK2/CDK2/BCL2L1	GADD45B
pss03460	Fanconi anemia pathway	0.03042	POLK/RAD51/BRCA2	
pss03440	Homologous recombination	0.03925	TOPBP1/RAD51/BARD1/BRCA2	

3.6.3. Changes in the immunity of *P. sinensis* under the influence of CHMs

CHMs mainly affects three immune-related signaling pathways: the FoxO signaling pathway (pss04068), PPAR signaling pathway (pss03320), and cytokine and cytokine receptor interaction (pss04060). The FoxO signaling pathway (pss04068) plays a role in regulating the cell cycle and controlling programmed cell death (Zhou et al., 2020a). Among them, it is worth mentioning that the signal transducer and activator of the transcription 3 gene (STAT3) were down-regulated in the experimental group (Fig. 6B). This plays a regulatory role in chronic inflammation causing tumors, and STAT3 deletion significantly inhibits tumorigenesis (Ashrafzadeh et al., 2021).

Peroxisome proliferator-activated receptors (PPARs), a subfamily of nuclear receptors, have been reported to regulate lipid metabolism and inflammatory responses, and they may signal dietary lipid uptake to the level of immune regulation (Varga et al., 2011). The transcriptome results of this experiment showed that four genes were up-regulated and three genes were down-regulated in the PPAR pathway, with the reported important gene HMGCS1 (3-hydroxy-3-methylglutaryl-CoA synthase 1 gene) being down-regulated (Table 4). HMGCS1 is an important enzyme that promotes cholesterol de novo synthesis, and inhibition of HMGCS1 reduces colon cancer cell proliferation (Zhou et al., 2020b).

Different factors in the cytokine and cytokine receptor interaction (pss04060) pathway play different roles. Among them, the expression of IL1R2 (interleukin 1 receptor type 2), CCL5 (C-C motif chemokine ligand 5), and CCL16 (C-C-motif chemokine ligand 16 genes) were down-regulated (Table 4). Down-regulation of these three genes can inhibit tumors (Jin et al., 2021; Shen et al., 2021b; Liu et al., 2022;). Notably, there are genes in other pathways that are also associated with the immunity, such as the down-regulated expression of prolyl 4-hydroxylase subunit alpha 1 (P4HA1) gene and activated T cell nuclear factor 3 (NFATc3) gene in Fig. 6. Down-regulation of P4HA1 promotes M1 microglial polarization and inhibits glioma invasion (Wang et al., 2021b). NFATc3 is a Ca²⁺-dependent transcription factor closely associated with the development of pulmonary arterial hypertension (PAH) (Zhu et al., 2015).

In addition, the results of the immunological indicators showed changes in LZM content, concentrations of C3 and IL-6, and activity of ACP (Fig. 2). In the transcriptome sequencing results, the changes in the expression of IL-6 related genes and ACP related genes (ACP7, ACP6, and ACP2) in the experimental group were also consistent with the results measured in this experiment. In sum, the results of the index measurement are consistent. The results showed that the immune capacity of *P. sinensis* was improved under the influence of the CHMs.

3.6.4. Effects of Chinese herbal medicine on growth and metabolism of *P. sinensis*

The cell cycle is closely related to cell proliferation, a fundamental feature of life, and is associated with essential body functions such as ontogeny. There are 20 genes up-regulated and 1 gene down-regulated in the cell cycle, among which the important genes worth mentioning are ORC3 (origin recognition complex subunit 3), CDK2 (cyclin-dependent kinase 2), CDK6 (cyclin-dependent kinase 6), E2F1 (E2F transcription factor 1) are up-regulated and GADD45B (growth arrest and DNA damage-inducible beta) is down-regulated. Together they promote cell proliferation (Fig. 6C) (Arellano et al., 1997; Bhattacharya et al., 2014). The GADD45B triggers growth arrest and DNA damage and plays a role in cell growth (Liu et al., 2009). The down-regulation of GADD45B in this experiment indicated that CHMs may be able to promote the growth of *P. sinensis*.

Four metabolism-related pathways, arginine and proline metabolism (pss00330), terpenoid backbone biosynthesis (pss00900),

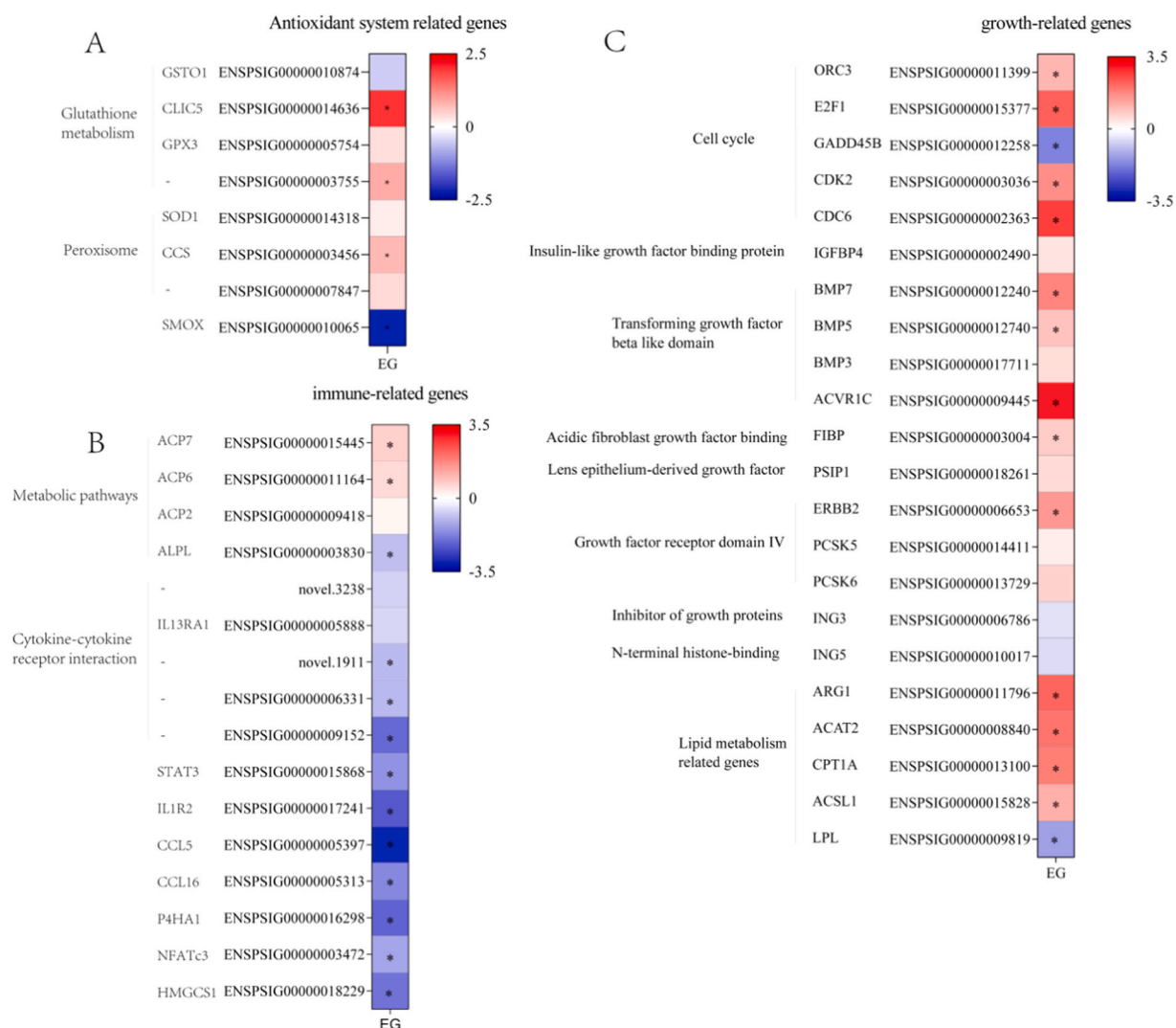


Fig. 6. Antioxidant system, immune-related genes, and growth-related gene expression.

primary bile acid biosynthesis (pss00120), and adipokine signaling pathway (pss04920), were identified. Among them, arginine is an essential amino acid that regulates feed intake and influences weight in animals (Hoseini et al., 2020). Notably, the arginase 1 gene (ARG1) was up-regulated in this experiment (Fig. 6C), and its up-regulation can lead to weight gain (Ogino et al., 2021).

The adipocytokine signaling pathway (pss04920) regulates energy intake, consumption, glucose and lipid metabolism, and anti-inflammatory and pro-inflammatory effects through autocrine, paracrine, and endocrine methods (Zhou et al., 2022). ACSL1 (acyl-CoA synthetase long-chain family member 1), CPT1A (carnitine palmitoyltransferase 1A), and ACAT2 (acetyl-CoA acetyltransferase 2) were up-regulated and LPL (lipoprotein lipase) was down-regulated. Together they affected lipid metabolism (Fig. 6C). Both up-regulation of ACSL1 and down-regulation of LPL increase triglyceride levels in vivo, and CPT1A and ACAT2 have a role in balancing body metabolism (Li et al., 2009; Zhang et al., 2022).

In addition, activin A receptor type 1C (ACVR1C) and bone morphogenetic protein 7 gene (BMP7) are up-regulated as growth factors in the cytokine and cytokine receptor interaction (pss04060) pathway (Fig. 6C), affecting the growth and metabolism in *P. sinensis*. ACVR1C, also known as the activin receptor-like kinase 7 gene (ALK7), is a type I receptor of the transforming growth factor beta (TGF- β) superfamily and is also involved in lipid metabolism by inhibiting lipolysis to promote lipid accumulation (Yogosawa et al., 2013). The BMP7 regulates bone formation, and it has been reported to enhance osteogenic differentiation of bone marrow mesenchymal stem cells (Asila et al., 2021). In sum, it is speculated from the transcriptome results that CHMs may improve the growth performance of *P. sinensis* by affecting lipid metabolism.

4. Conclusion

This study investigated the effects of CHMs on the growth performance and immune responses of the Chinese soft-shelled turtle (*P. sinensis*) in greenhouse mode. The compound CHMs can increase the weight gain rate and the immune protection rate against *A. hydrophila* of *P. sinensis*. After feeding with CHMs supplemented diet for 60 d, the antioxidant system and immunological indexes of *P. sinensis* changed significantly. In addition, comparative transcriptome analysis revealed that growth, immunity, and lipid metabolism-related pathways are involved in the response to CHMs. Among them, two pathways—cell cycle and cytokine-cytokine receptor interaction—may play important roles in the growth and immunity of *P. sinensis*. SMOX, STAT3, and HMGCs1 may be the key genes of CHMs affecting the immune response of *P. sinensis*. GADD45B, ARG1, ACVR1C, and BMP7 may be key genes affecting growth performance. This study provides a theoretical and practical basis for the green and efficient culture of *P. sinensis*.

CRedit authorship contribution statement

Yuqi Mu: Investigation, Validation. **Yaguang Wang:** Data curation, Methodology, Project administration, Writing – review & editing. **Mengyan Lan:** Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Yueqiang Guan:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing. **Zhao Zhang:** Conceptualization, Project administration, Supervision, Writing – review & editing. **Yali Li:** Investigation, Validation.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.anifeedsci.2024.116084](https://doi.org/10.1016/j.anifeedsci.2024.116084).

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